



EXPLANATION

Informative companion to OWL-SEMAPHORE-SYSTEM

“Four owls tell the reader what kind of thinking they are looking at.”

Informative (this entire document)

Owl Semaphore — Explanation

Origin story, archetype rationale, why two states were not enough

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SOURCE github.com/IT-Help-San-Diego/owl-semaphore VERSION v1.3.0-rc · LICENSE CC BY 4.0

Canonical: A finite algebra over epistemic states, implemented as a reproducible visual notation system with enforced invariants.

```
BANNER TUPLE BEGIN
state=EXPLANATION
transform=I
determinant=+1
mapping=(x, y) → (x, y)
version=v1.3.0-rc
concept_doi=10.5281/zenodo.19473697
version_doi=TBD_BY_ZENODO_ON_RELEASE
last_published_doi=10.5281/zenodo.19474599
license=CC BY 4.0
BANNER TUPLE END
```



OWL SEMAPHORE — FOUR-STATE MASTER PROOF

OWL SEMAPHORE SYSTEM

Klein Four-Group V4 = {I, sigma-v, C2, sigma-h} | Epistemic State Notation | 1080px Masters

1 NORMATIVE

$T = I \quad \det = +1 \quad (x, y) \rightarrow (x, y)$




Field: (15,15,15)
Ring: (212,168,83)

Upright, faces RIGHT
Identity / Canonical / Baseline
Binding methodology, reference standard

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
Contrast: 8.8:1
Gold on black
4 layers + composite | 1080x1080 RGBA

2 NON-NORMATIVE

$T = \text{sigma-v} \quad \det = -1 \quad (x, y) \rightarrow (-x, y)$




Field: (213,175,100)
Ring: (44,93,88)

Upright, faces LEFT
Alternative / Counter-perspective
Legitimate dissent, reframing

$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$
Contrast: 3.8:1
Teal on gold
5 layers + composite | 1080x1080 RGBA

3 CRITICAL

$T = C2 \quad \det = +1 \quad (x, y) \rightarrow (-x, -y)$




Field: (43,45,44)
Ring: (151,16,30)

Inverted, faces LEFT
Falsification / Total inversion
Red team, adversarial analysis

$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$
Contrast: 1.3:1
Red on dark gray
5 layers + composite | 1080x1080 RGBA

4 METACOGNITIVE

$T = \text{sigma-h} \quad \det = -1 \quad (x, y) \rightarrow (x, -y)$




Field: (26,16,32)
Ring: (140,65,145)

Inverted, faces RIGHT
Self-examination / Frame audit
The observer audits the observer

$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Contrast: 2.4:1
Amethyst on violet-black
5 layers + composite | 1080x1080 RGBA

Owl Semaphore System | V4 subset O(2) | SHA-3-512 verified | DOI: 10.5281/zenodo.18854899
sigma-v, sigma-h = C2 | sigma-v, C2 = sigma-h | sigma-h, C2 = sigma-v | All elements self-inverse

- Concept DOI (all versions): 10.5281/zenodo.19473697
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- Version DOI (v1.3.0): TBD_BY_ZENODO_ON_RELEASE
- License: CC BY 4.0



This document is the long-form companion to `OWL-SEMAPHORE-SYSTEM.md`. The system specification is the formal definition; this document is the public-readable origin story, the design rationale, and the bridge to the DNS Tool research context where the notation was first needed.

This entire document is **informative**, in the ISO/IEC Directives Part 2 sense. It places no interoperability requirement on implementations. The normative material lives in the system specification and the four state specifications.

1. Canonical wording stack

The Owl Semaphore is described at three different layers, all of which point to the same object. Use whichever layer is right for the reader.

Layer	Sentence	Use
Formal	A finite algebra over epistemic states, implemented as a reproducible visual notation system with enforced invariants.	README, system specification, Zenodo metadata, citation abstract.
Operational	A four-state visual system for marking how a claim, document, dataset, or finding should be evaluated before belief, challenge, or action.	This document, the DNS Tool website bridge, public overview.
Human	Four owls tell the reader what kind of thinking they are looking at: standard, exploration, inversion, or self-audit.	Story sections, introductions, teaching material.

The four states named in the Human sentence map one-to-one onto the four formal states (NORMATIVE $\rightarrow$ standard; NON-NORMATIVE $\rightarrow$ exploration; CRITICAL $\rightarrow$ inversion; METACOGNITIVE $\rightarrow$ self-audit).

2. Why the Owl Semaphore exists

The Owl Semaphore was not designed as a decorative badge set. It was designed inside the DNS Tool project (`dnstool.it-help.tech`) to solve a specific recurring problem: when a research tool is read by experts from very different cultures, those cultures fail to agree about what they are looking at long before they fail to agree about the content.

DNS Tool was built around the idea that the internet needs research tools that are technically deep, culturally legible, and operationally respectful. It separates observed facts from analytical inference, hashes results for tamper-evidence, uses resolver consensus, reports calibrated uncertainty, and follows a minimal-footprint good-net-citizen scanning posture (see DNS Tool



confidence). It documents methodology, philosophical foundations, communication standards, rules of engagement, architecture, sources, and citations as distinct governance artefacts (see DNS Tool publications). It already uses the Owl Semaphore as an epistemic classification legend across its corpus (see DNS Tool corpus).

That work surfaced the missing piece: a visual-algebra layer that lets a reader see, before they engage the content at all, what kind of evaluation is being requested.

3. The archetype problem

DNS Tool's first readers came from at least seven overlapping expert cultures, each with its own failure mode when a tool fails to respect its standards. The Owl Semaphore was designed to honor every one of these cultures at once without flattening any of them.



Archetype	What they demand	What the system gives them
Hackers / red-team / offensive operators	Proof. Reproducibility. No vendor incantations.	An explicit transform, determinant, mapping, and SHA-3-512 hash per artefact.
OSINT / intelligence analysts	Source discipline; separation of assertion from analysis; calibrated language.	A METACOGNITIVE state and an audit register distinct from NORMATIVE/CRITICAL. References ICD 203 tradecraft.
DNS / protocol engineers	Protocol correctness. Reproducible behavior. Honest error states.	A NON-NORMATIVE state distinct from CRITICAL so that “this departs from the standard” never collapses into “this attacks the standard”.
RFC / standards readers	BCP 14 normative discipline. Uppercase MUST kept rare and meaningful.	RFC 2119 / RFC 8174 BCP 14 boilerplate; uppercase keywords restricted to genuine interoperability requirements.
Data scientists	Reproducible builds. Determinism. Provenance.	Single-command PDF regeneration (<code>make pdfs</code>), integrity manifest, hash coverage, banner-tuple test from the rendered output.
Serious operators	Don’t waste my time, don’t lie about uncertainty, don’t give me theatre.	Color is never the only channel; redundant encoding through color + orientation + label.
Public readers	Comprehensible truth without dumbing down.	The Human canonical sentence: “Four owls tell the reader what kind of thinking they are looking at.”

The insight is that a research tool can be deep, rigorous, and culturally legible at the same time — but only if the cultural-legibility layer is itself formal. The Owl Semaphore is that layer.

4. Why two states were not enough



Standards culture is comfortable with a two-state distinction: **normative** (binding) and **non-normative** / **informative** (advisory). That distinction is genuinely useful and it is preserved here. But it was not enough for full epistemic congruence in DNS Tool research.

Two-state classification cannot distinguish, for example:

- “*This reflects the standard from a different angle*” (exploration) from “*This attacks or inverts the standard*” (inversion). Both are “non-normative” under a two-state scheme, but they are very different operational signals.
- “*The author audited the frame itself*” (self-audit) from “*The author proposed an alternative*” (exploration). Again, both collapse into “non-normative” with no further structure.

A claim can be authoritative, exploratory, inverted by its own evidence, or requiring an audit of the observer or frame. Four states are required to express this distinction without leaving the reader to guess which one applies.

5. Why V_4 (the Klein four-group)

Four states are the minimum that admit closure, reversibility, and a non-trivial composition law without collapsing into a linear order. The Klein four-group $V_4 = \{I, \sigma_v, C_2, \sigma_h\}$ is the unique abelian non-cyclic group of order 4 (Wikipedia; nLab). It has three useful properties for an epistemic-state notation:

1. **Closure.** Every composition of two states is again one of the four states. There is no “fifth mood” the system can drift into. This is established explicitly via the Cayley table in `OWL-SEMAPHORE-SYSTEM.md §2.3`.
2. **Reversibility.** Every state is its own inverse: applying any transform twice returns to **NORMATIVE**. There is no one-way decay path.
3. **Geometric realizability.** V_4 is exactly the symmetry group of a non-square rectangle in the plane: {identity, horizontal reflection, vertical reflection, 180° rotation}. This makes V_4 a finite subgroup of the orthogonal group $O(2)$ — embeddable as 2×2 diagonal matrices with entries in $\{+1, -1\}$ — which is exactly what is needed to map states to visual orientations.

Note the distinction: V_4 is a finite group; $O(2)$ is a continuous Lie group. The system uses V_4 as a finite subgroup of $O(2)$. It does not claim that V_4 is $O(2)$, and it does not claim that group structure by itself implies security, correctness, or analytic validity of the marked content. Group structure guarantees algebraic closure of the notation; the content still has to stand or fall on its own merits.

6. Why the owl

The Owl of Athena carries a long human association with wisdom, vigilance, and night-time perception. The choice here is not pure symbolism: in this system the owl is constrained by V_4



rather than used as decorative shorthand. The four owls share a common center, a common annular geometry, and a common meander ring; the only variation between them is the transform applied. The archetype is admitted; the math then disciplines it.

The four owls are an **iconic** sign (in Peirce’s sense) for the state itself — each posture *resembles* the operation it represents (upright/right-facing for identity; upright/left-facing for vertical reflection; upside-down/left-facing for rotation; upside-down/right-facing for horizontal reflection). The text labels (NORMATIVE / NON-NORMATIVE / CRITICAL / METACOGNITIVE) are **symbolic** signs (conventional, must be learned). The triple redundancy of color + orientation + label is what allows the system to remain readable in grayscale, under color vision deficiency, and in low-resolution rendering — see `OWL-SEMAPHORE-SYSTEM.md` §13 and WCAG 2.2 SC 1.4.1.

7. How the four states map to evaluation

Owl	State	Operator	Determinant	What the reader is looking at	DNS Tool corpus mapping
1	NORMATIVE	I	+1	The standard, the baseline framework, the binding reference.	“This is the standard.”
2	NON-NORMATIVE	σ_v	−1	A structured exploration of the standard, a legitimate alternative interpretation.	“This reflects the standard.”
3	CRITICAL	C_2	+1	An inversion of the standard’s assumptions; adversarial analysis.	“This inverts the standard.”
4	METACOGNITIVE	σ_h	−1	An audit of the observer or evaluative frame itself, not of the object.	“This audits the standard.”

The METACOGNITIVE state is the easiest one to misread. It does **not** mean “introspection about feelings”. It means *the observer audited the observer*: the object of analysis did not change, the observer did not change, but the evaluative frame was deliberately inverted. The canonical embodied analogy in `OWL-1-NORMATIVE.md` §20 is the through-the-legs maneuver — searching a room normally, failing to detect the target, then bending over and viewing the same room upside-down



through the legs. The room is unchanged. The frame is inverted. That is the METACOGNITIVE state in its purest physical form.

8. What the Owl Semaphore does not claim

A formal notation is only as useful as its boundaries. The following are **not** claims this system makes:

- It does **not** claim that group structure implies security, correctness, or analytic reliability of the marked content. Group structure guarantees closure of the notation, not validity of the conclusions.
 - It does **not** claim that NORMATIVE content is metaphysically true, immune to revision, or beyond challenge. It claims only that the content carries the highest internal obligation level inside this system.
 - It does **not** claim formal verification in the seL4 sense (see Klein et al., SOSP 2009; seL4 whitepaper). seL4 is referenced as an aspirational discipline standard for “explicit invariants and machine-checkable transition rules”; it is not evidence that this repository is formally verified.
 - It does **not** claim WCAG 2.2 Level AA conformance has been empirically verified. It claims the design *targets* AA conformance through redundant encoding; empirical conformance still requires testing.
 - It does **not** combine confidence statements and likelihood statements in the same sentence. Per ICD 203 these are orthogonal dimensions and the Owl Semaphore is a state notation, not a probability/confidence scale.
 - It does **not** claim that the $\sim 31^\circ$ rotation observed in earlier teal artwork is a fifth state. It is a process operator, not part of the closed V_4 state set.
-

9. How this connects to DNS Tool

DNS Tool’s confidence engine (dnstool.it-help.tech/confidence) already separates observed facts from analytical inference, hashes results, uses resolver consensus, reports uncertainty, and distinguishes success / rate-limited / error / partial states. The Owl Semaphore gives that discipline a public-facing visual grammar. The DNS Tool corpus (dnstool.it-help.tech/corpus) uses the four owls as the legend for classifying audited standards and findings. The DNS Tool publications page (dnstool.it-help.tech/publications) treats the Owl Semaphore concept DOI as a peer artefact alongside methodology, philosophical foundations, communication standards, rules of engagement, architecture, and the confidence framework.

10. How this connects to the larger Intellectual Resistance framework



The Owl Semaphore sits inside a fourfold operating field developed in parallel with the DNS Tool research program:

1. **Verification Principle** — verify the claim.
2. **Carrier Color Theory** — clean the carrier.
3. **Symbiotic Net Citizen Principle** — protect the system.
4. **Versioning and Provenance Rule** — preserve the path.

In that field, the Owl Semaphore marks the **state of thought before action**. It is the visible state layer that helps a research program maintain congruence across claim, carrier, code, consequence, and correction.

11. References

- DNS Tool Owl Semaphore
- DNS Tool confidence
- DNS Tool corpus
- DNS Tool publications
- RFC 2119 — Key words for use in RFCs to Indicate Requirement Levels
- RFC 8174 — Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words
- Klein four-group — Wikipedia
- Klein four-group — nLab
- Knill, “Unit 8: The orthogonal group,” Harvard Math 22b notes
- WCAG 2.2 — W3C Recommendation
- WCAG 2.2 SC 1.4.1 — Use of Color
- Section 508 — Making Color Usage Accessible
- PMC — Global Perspective of Color Vision Deficiency
- Moody 2009 — Designing Effective Visual Notations
- ICD 203 — Analytic Standards
- Klein et al. 2009 — seL4: Formal Verification of an OS Kernel
- seL4 Whitepaper



OWL SEMAPHORE SYSTEM — CLASSIFICATION LEDGER



NORMATIVE

$T = I \quad \det = +1$

$(x, y) \mapsto (x, y)$

“This is the standard.”

RFC 2119 MUST / SHALL



NON-NORMATIVE

$T = \sigma_v \quad \det = -1$

$(x, y) \mapsto (-x, y)$

“This reflects the standard.”

Informative / Advisory (NOTE)



CRITICAL

$T = C_2 \quad \det = +1$

$(x, y) \mapsto (-x, -y)$

“This inverts the standard.”

RFC 2119 MUST NOT / SHALL NOT



METACOGNITIVE

$T = \sigma_h \quad \det = -1$

$(x, y) \mapsto (x, -y)$

“This audits the standard.”

Epistemic / Framework (META)

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